

Simple Smart ELF Loader Documentation

Group 4

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1 Introduction

This document provides documentation for the Simple Smart ELF Loader written in C. The program is designed to load and execute ELF files, handling segmentation faults and memory management.

2 Code Explanation

2.1 Global Variables

```
Elf32_Ehdr *elf_header; // ELF header
int file_descriptor; // File descriptor
int system_page_size; // System page size
int page_fault_count = 0;
int page_allocation_count = 0;
float total_internal_fragmentation = 0.0f;
```

These global variables store information about the ELF file, system page size, and various statistics related to memory management and segmentation faults.

2.2 Signal Handler

```
void segmentation_fault_handler(int signal_number, siginfo_t *signal_info, void
    *context) {
    // ...
}
```

This function is the signal handler for segmentation faults. It attempts to allocate memory for the faulting address within a PT_LOAD segment, loads the segment data, and updates statistics. If the address doesn't belong to any segment, it reports an invalid memory access.

2.3 Main Function

```
int main(int argc, char** argv) {
    // ...
}
```

The 'main' function sets up the signal handler, opens the ELF file, reads the ELF header, and attempts to execute the '_start' method. It also prints the entry point address and reports statistics.

3 Compile and Run

To compile and run the Simple Smart ELF Loader, follow these steps:

```
$ make
$ ./loader given_file
```

Replace 'given_file' with the path to the ELF executable you want to load.

4 Contribution

Gagan's Contribution:

Handled the setup of the code, including setting up the signal handler, opening the ELF file, reading the ELF header, managing the initial setup and executing the '`_start`' method

Garv's Contribution:

Garv's contribution focused on managing the cleanup process, printing the entry point address, and reporting statistics on page faults and internal fragmentation.

5 Conclusion

The Simple Smart ELF Loader is a versatile tool for loading and executing ELF executables while managing memory and handling segmentation faults.